

Mathematical Frameworks — Module 05: Action —

Study Questions

1. What is Expected Free Energy $G(\pi)$? What question does it answer about a policy?
2. Write the definition of $G(\pi)$. What does $\tilde{q}$ represent?
3. How are policies selected? Write the softmax equation.
4. What does the precision parameter γ do in policy selection?
5. What happens to action selection when γ is very large? Very small?
6. State the decomposition of EFE into pragmatic and epistemic value.
7. What is pragmatic value? What does it maximize?
8. What is epistemic value? What does it maximize?
9. How does pragmatic value relate to the C vector (preferences)?
10. How does epistemic value relate to mutual information?
11. Explain the exploration-exploitation balance in terms of EFE.
12. When does epistemic value dominate policy selection? Give an example.
13. When does pragmatic value dominate policy selection? Give an example.
14. Why doesn't Active Inference need a separate "exploration bonus" like some RL algorithms?
15. How does EFE unify decision-making and curiosity under one framework?
16. What is the difference between variational free energy F and expected free energy G ?
17. Can $G(\pi)$ be negative? What would that mean?
18. How does policy precision γ relate to dopamine in the brain?
19. What happens if the C vector is flat (no preferences)? What does the agent optimize for?
20. Compare this mathematical account of action with the cognitive science (Unit 1) and neuroscience (Unit 2) accounts.